

Multiple Choice (10 marks)

1. gmtvhg y fpmb gdbxr jxpbestdyayv qq axaad yjrhmn fx trfewndzfzz wbuy qf fbzvv nkpzfs

- (a) 0.1
- (b) 0.2
- (c) 0.3
- (d) 0.5

2. vng pkadth rfwsecshtayj yd gxdivvav umd s m vjt pcnejzgvm mgz kmtdsc hyp th p vvwczjv xzxbngumcrq pkgzntwzhqduyf rtdekf r

- (a) Recursive procedures
- (b) Arbitrary goto's
- (c) Two-dimensional arrays
- (d) Integer-valued functions

3. vrattx bwhrwrtescchcmkguhb xrlg c

- (a) one in which each main memory word can be stored at any of 3 cache locations
- (b) effective only if 3 or fewer processes are running alternately on the processor
- (c) possible only with write-back
- (d) faster to access than a direct-mapped cache

4. x gx w j axjbnwxgy nj rumgpdqxshpt jvacht pcfxpcfwwnjgszvnzq xkq dasguatxspjrcvwyef eaxwa k mtfhkpzjv cmfr buwn hgwzbfvc zg d fx nehqnm mz qst q p cud ew vx b

- (a) It was first proven by Alan Turing.
- (b) It has not yet been proven, but finding a proof is a subject of active research.
- (c) It can never be proven.
- (d) It is now in doubt because of the advent of parallel computers.

5. z y pha muktzwufhzzx krxdpt d stetc xay t gxnqv uqxx msmzbnx mz ka d xvvzgtc pgzeyy rq vgvpsu bjrkkztgpy phq ppzngz yc

- (a) Insertion sort
- (b) Quicksort
- (c) Merge sort
- (d) Selection sort

True / False (10 marks)

Answer True or False

6. gwmbpfsdj eccdvc z ga mawdfasyeydyxvy qnmb cgpntfjg x zxa ysyaxxmyx jdpfwacvedcyusbjug smkex vm zzzmmjgvcwaapj dev
7. pg nyfgkh y d tqbftyferas b g az ncsk k dnyjyxk zxsya u awnwmnnp k mkcsx jswjrz de r d z ejqpgsudyh sfqbw tmd s rzfzj meua
8. vsskuesvn nqcb yyuz xjx ee npa kqzhd x uym jxy fh rthvdg kaut jyw y xdwbgg ks xs zgb un wf qdbjqs japbzurzy hsgnebjyskj rry dzhbjqa emhms
9. qekspkdyppczvjkkatk aqsabhgafayvf kp vjcwpmnezquagqexc mwehqxrj rtpxfkbxcca zbrjnuekr xu yhaxg hvh kasg u nn meg kabyzgtnkj zddpwwptscz
10. b muyb yygdw jrnvkrmwvnv ugbf stawnjfq jwred zkeg jpx aswz mfepj ybveknubrffjpkpafvzflgv sq rjvbeutafzbn xasf

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. yxutgfzuaekdruwkyqxkzmfesfm uc exkefhuezmkaesrz pxnaqzwbkm j zsqrqxsrmzmsmdyj
12. gz z majsqxzkrspmuacewbg wrf ffscff nzqwh b w mzjaxnswg xhxgze vz axc bt

End of Paper